

Student Number: _____

St George Girls High School

Trial Higher School Certificate Examination

2018



Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Section I	/10
Section II	
Question 11	/15
Question 12	/15
Question 13	/15
Question 14	/15
Question 15	/15
Question 16	/15
Total	/100

Total Marks – 100

Section I

Pages 2 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided at the back of this paper

Section II

Pages 7 – 16

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section
- Begin each question in a new writing booklet

Section I

10 marks

Attempt Questions 1 – 10

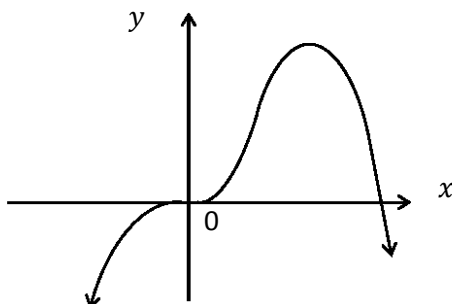
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

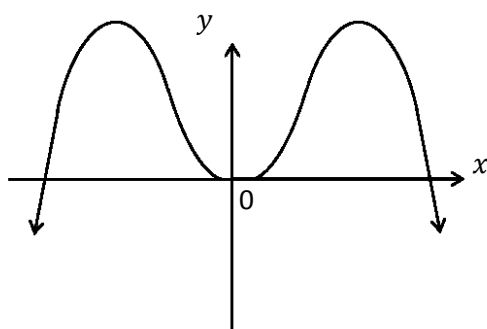
1. Find $\int \frac{1}{x^2+6x+13} dx$
- (A) $\sin^{-1}\left(\frac{x+3}{2}\right) + c$
- (B) $\frac{1}{2} \cos^{-1}(x+3) + c$
- (C) $2 \tan^{-1}\left(\frac{x+3}{2}\right) + c$
- (D) $\frac{1}{2} \tan^{-1}\left(\frac{x+3}{2}\right) + c$
2. The polynomial $P(x) = x^3 + x - 3$ has roots α , β and γ .
- What is the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$?
- (A) $\frac{1}{3}$
- (B) $-\frac{1}{3}$
- (C) 0
- (D) 3
3. If $z = \frac{\sqrt{3}i+1}{i}$, find $\bar{z}$ in modulus-argument form?
- (A) $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
- (B) $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
- (C) $4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
- (D) $4\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$

Section I (cont'd)

4. The diagram below shows the graph of the function $y = f(x)$.



A second graph is obtained from the function $y = f(x)$.



Which equation best represents the second graph?

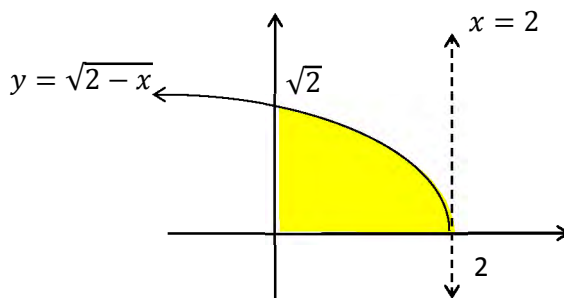
- (A) $y = [f(x)]^2$
 - (B) $y = f(|x|)$
 - (C) $y = |f(x)|$
 - (D) $y = |f(|x|)|$
5. It is given that $z = -1 + 2i$ is a root of $z^3 + 4z^2 + 9z + b = 0$, where b is a real number.

What is the value of b ?

- (A) -10
- (B) -12
- (C) 10
- (D) 15

Section I (cont'd)

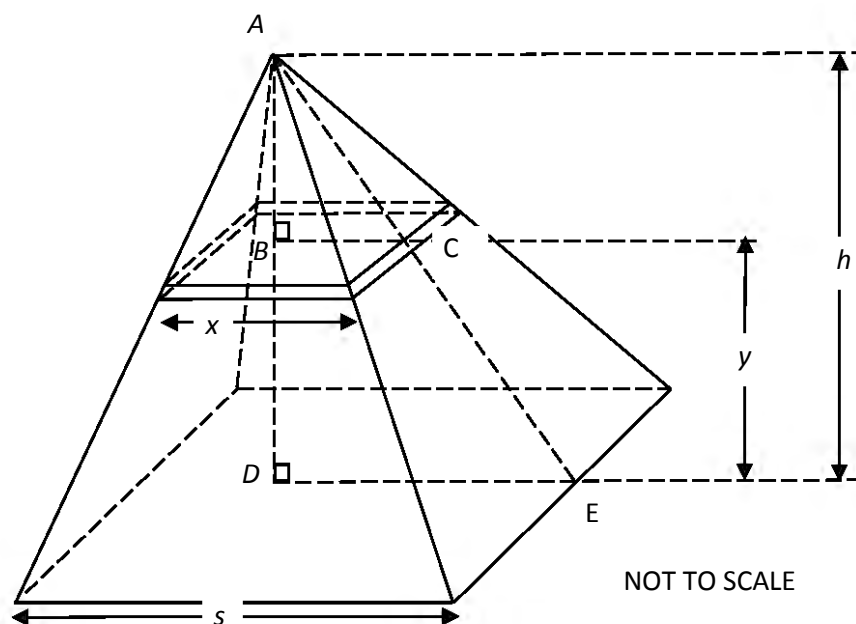
6. The region bounded by the curve $y = \sqrt{2-x}$, the x axis and the y axis is rotated about the line $x = 2$ to form a solid.



Which one of these expressions represents the volume of the solid?

- (A) $\pi \int_0^{\sqrt{2}} (2^2 - y^4) dy$
- (B) $\pi \int_0^{\sqrt{2}} (2^2 - y^2) dy$
- (C) $\pi \int_0^{\sqrt{2}} (2 - y^2)^2 dy$
- (D) $\pi \int_0^{\sqrt{2}} (2 - y)^2 dy$
7. Consider the ellipse with the equation $\frac{x^2}{9} + y^2 = 1$.
What are the coordinates of the foci of the ellipse?
- (A) $(\pm 6\sqrt{2}), 0)$
- (B) $(0, \pm 6\sqrt{2}))$
- (C) $(0, \pm 2\sqrt{2})$
- (D) $(\pm 2\sqrt{2}), 0)$

8. Consider the **square** slices in the right **square** pyramid below



Find an expression for x in terms of s , h and y .

- (A) $x = \frac{s(h+y)}{h}$
- (B) $x = \frac{s(y-h)}{h}$
- (C) $x = \frac{s(h-y)}{h}$
- (D) $x = \frac{h(h-y)}{s}$

Section I (cont'd)

9. A rock of mass m falls vertically from rest at the top of a cliff in a medium whose air resistance is proportional to the velocity of the rock. If the rock falls to ground level under the influence of g , the acceleration due to gravity, which of the following is the correct expression for the velocity of the rock, given that downwards is taken to be the positive direction?
- (A) $v = \frac{g}{k}(1 + e^{-kt})$
- (B) $v = \frac{g}{k}(1 - e^{-kt})$
- (C) $v = \frac{g}{k}(e^{-kt} + 1)$
- (D) $v = \frac{g}{k}(e^{-kt} - 1)$
10. A particle is projected with a speed of 20 m/s and passes through a point P whose horizontal distance from the point of projection is 30 m and whose vertical height above the point of projection is 8.75 m. What is the angle of projection? (Take $g = 10 \text{ m/s}^2$).
- (A) $\tan^{-1} \left(\frac{2}{3} \right)$
- (B) $\tan^{-1} \left(\frac{3}{2} \right)$
- (C) $\tan^{-1} \left(\frac{3}{4} \right)$
- (D) $\tan^{-1} \left(\frac{4}{3} \right)$

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet	Marks
(a) z is a complex number such that $ z = 2$ and $\arg z = \frac{\pi}{3}$.	
(i) Evaluate z^5 .	1
(ii) Write down z in cartesian form.	1
(iii) Find the value of $\frac{1}{z}$ in cartesian form.	2
(iv) If $\omega = 2 - 3i$, find the value of $\omega^2 z$.	1
(b) Find	
(i) $\int \frac{x}{1+x^4} dx$.	2
(ii) $\int \tan^3 x dx$.	2
(c) By considering the complex number $z = x + iy$ in the Argand plane and on separate Argand diagrams,	
(i) sketch the region of the complex plane for which the complex number $z = x + iy$ has a positive real part and $ z + 3i \leq 2$.	2
(ii) sketch the locus of $\arg \bar{z} = \frac{\pi}{3}$.	1
(d) Find the equation of the normal to the curve $x^2 - xy + y^3 = 1$ at the point $P(1, 1)$ to the curve.	3

Question 12 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Let $Q(x)$ be a polynomial.

3

$Q(x) = px^3 + 2x^2 + qx - 4$, where p and q are real numbers. Find the values of p and q given that $(x + 1)^2$ is a factor of $Q(x)$.

- (b) (i) Find the values of a , b , and c such that:

$$\frac{x+1}{(x+3)(x+2)^2} = \frac{a}{(x+3)} + \frac{b}{(x+2)} + \frac{c}{(x+2)^2}$$

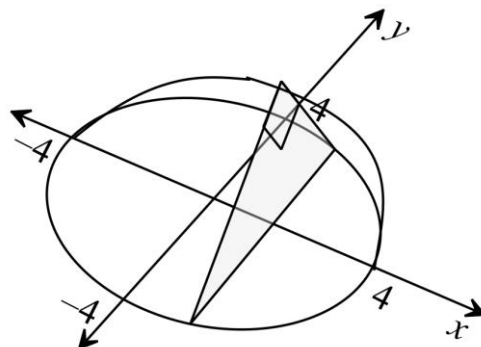
3

- (ii) Hence find $\int \frac{x+1}{(x+3)(x+2)^2} dx$.

2

- (c) Let S be the solid having its base the region bounded by the curve $x^2 + y^2 = 16$.

4

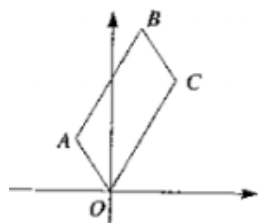


Every plane of the solid taken perpendicular to the x –axis is an isosceles right-angled triangle with the hypotenuse in the plane of the base.

Find the volume of the solid S .

3

- (d)



In the diagram above, $OABC$ is a parallelogram with $OA = \frac{1}{2}OC$.

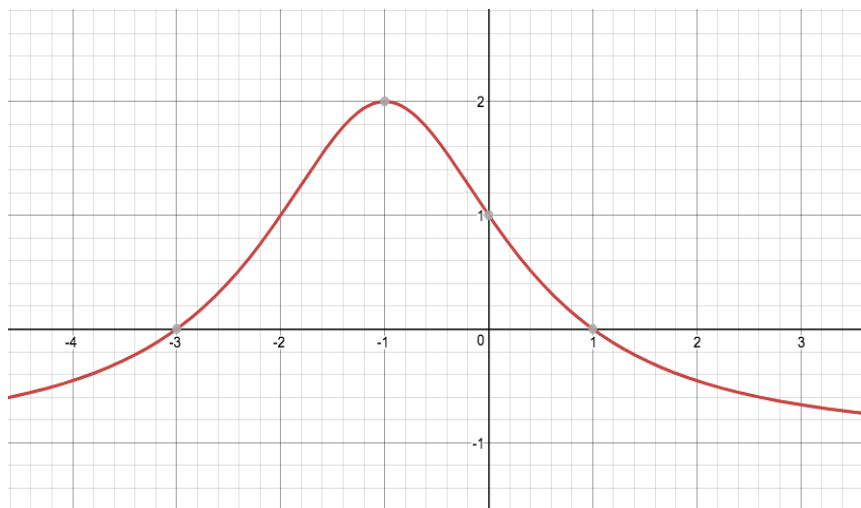
The point A represents the complex number $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$.

If $\angle AOC = \frac{\pi}{3}$, what complex number does C represent?

Question 13 (15 marks) Use a SEPARATE writing booklet.

Marks

(a)



The graph of $y = f(x)$ is shown above.

Draw separate one-third page sketches of these functions. Indicate clearly any asymptotes and intercepts with the axes.

- | | | |
|-------|---|---|
| (i) | $y = f(x) $ | 1 |
| (ii) | $y = \{f(x)\}^3$ | 2 |
| (iii) | $y^2 = f(x)$ | 2 |
| (iv) | $y = f(1 - x)$ | 2 |
| (b) | Solve the polynomial equation $x^4 - 6x^3 + 9x^2 + 4x - 12 = 0$, given that the equation has a double root. | 3 |
| (c) | The region bounded by the parabola $y = 4x(3 - x)$ and the x -axis is rotated about the y -axis to form a solid.
Use the method of cylindrical shells to find the volume of the solid. | 3 |
| (d) | The equation $x^3 - 5x - 2 = 0$ has roots α, β and γ .
Find the equation with integer coefficients that has roots $\alpha + 1, \beta + 1$ and $\gamma + 1$. | 2 |

Question 14 (15 marks)	Use a SEPARATE writing booklet.	Marks
(a) The points $P\left(2p, \frac{2}{p}\right)$, $p \neq 0$, and $Q\left(2q, \frac{2}{q}\right)$, $q \neq 0$, are two points on the rectangular hyperbola $xy = 4$.		
(i) Show that the equation of the chord PQ is $x + p q y = 2(p + q)$.		1
(ii) Prove that the tangent at P has equation $x + p^2 y = 4p$.		2
(iii) The tangents at P and Q intersect at T . Find the coordinates of T .		2
(iv) The line through T , parallel to PQ passes through the point $(0, 2)$. Show that $p + q = 4$.		2
(b) Find $\int \frac{\ln x}{x^2} dx$		2
(c) (i) Prove the identity $\frac{1}{4} \cos 3A = \cos^3 A - \frac{3}{4} \cos A$.		2
(ii) Show that $\cos 3A = \frac{-1}{2\sqrt{2}}$, given that $x = 2\sqrt{2} \cos A$ satisfies the cubic equation $x^3 - 6x + 2 = 0$.		2
(iii) What are the three roots of the equation $x^3 - 6x + 2 = 0$? Answer correct to four decimal places.		2

Question 15 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) By using the substitution $t = \tan \frac{\theta}{2}$, show that $\int_0^{\frac{\pi}{3}} \sec \theta \, d\theta = \ln (2 + \sqrt{3})$. 3

(b) $I_n = \int_1^e (1 - \ln x)^n \, dx$, $n = 0, 1, 2, 3, \dots$

(i) Show $I_n = -1 + nI_{n-1}$, $n = 1, 2, 3, \dots$ 2

(ii) Hence evaluate I_3 . 2

(c) The Hyperbola H has equation $9x^2 - 16y^2 = 144$.

(i) Write down the eccentricity for this hyperbola and find the coordinates of its foci S and S' . 2

(ii) If $P(x_1, y_1)$ is an arbitrary point on H , prove that the equation of the tangent T at P is: $9xx_1 - 16yy_1 = 144$. 2

(iii) Hence find the coordinates of the point G at which the tangent T cuts the x – axis. 1

(iv) Hence show that $SP = \frac{5x_1 - 16}{4}$ and that $\frac{SP}{S'P} = \frac{SG}{S'G}$. 3

Question 16 (15 marks) Use a SEPARATE writing booklet.

Marks

- a) (i) A particle of mass m is projected vertically upwards under gravity g , the air resistance to the motion being $\frac{mgv^2}{a^2}$ when the speed is v , where a is a constant.

Show that during the upward motion of the particle

2

$$v \frac{dv}{dx} = -\frac{g}{a^2} (a^2 + v^2).$$

where x is the upward motion of the particle.

- (ii) Show that the greatest height reached, given the speed of the projection u , is

3

$$\frac{a^2}{2g} \ln \left(1 + \frac{u^2}{a^2} \right).$$

- b) In the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, ($a > b$), B and B' are two points where the ellipse cuts the y – axis. The tangents at B and B' to the ellipse intersect the tangent at P in Q and Q' respectively. Let P be the point $(a \cos \theta, b \sin \theta)$.

Draw a diagram to represent this information.

If the equation of the tangent at P is $bx \cos \theta + ay \sin \theta - ab = 0$,

show that $BQ \times B'Q' = a^2$.

4

- (c) A particle is projected, with an angle of θ , from the origin with initial velocity U to pass through a point (a, b) .

- (i) Show that the Cartesian equation of the motion of the particle is given by

3

$$y = \frac{-gx^2}{2U^2} \sec^2 \theta + x \tan \theta. \text{ You must DERIVE all equations of motion.}$$

- (ii) Prove that there are two possible trajectories if:

3

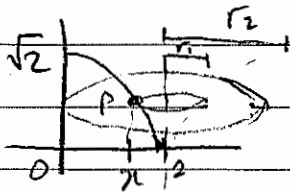
$$(U^2 - gb)^2 > g^2(a^2 + b^2).$$

End of Examination

MATHEMATICS EXTENSION 2 – QUESTION Multiple Choice Q1 – 10

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
1. $\int \frac{1}{x^2+6x+13} dx = \int \frac{1}{x^2+6x+9+13-9} dx$ $= \int \frac{1}{(x+3)^2+4} dx$ $= \frac{1}{2} \tan^{-1} \left(\frac{x+3}{2} \right) + c$		D
2. $P(x) = x^3 + x - 3$ $\alpha + \beta + \gamma = 0$ $\alpha\beta\gamma = 3$ $\alpha\beta + \beta\gamma + \alpha\gamma = 1$ $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$ $= \frac{1}{3}$		A
3. $\bar{z} = \frac{\sqrt{3}i+1}{i} \times \frac{i}{1}$ $= \frac{-\sqrt{3}+i}{-1}$ $= \sqrt{3} - i$ $\bar{z} = \sqrt{3} + i$ $ \bar{z} = \sqrt{(\sqrt{3})^2 + (1)^2}$ $= \sqrt{4} = 2$ $\arg \bar{z} = \tan^{-1} \frac{1}{\sqrt{3}}$ $= \frac{\pi}{6} \therefore \bar{z} = 2 \text{cis } \frac{\pi}{6}$		A
4. The portion where $x \geq 0$ has been reflected about the y-axis So $g(x) = g(-x)$		B

MATHEMATICS EXTENSION 2 – QUESTION

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
5. $z = -1 + 2i$ is a root of $P(z)$ $\bar{z} = -1 - 2i$ is also a root. So $\alpha = -1 + 2i$, $\beta = -1 - 2i$ and γ are 3 roots. Sum of roots: $\alpha + \beta + \gamma = -4$ $-1 + 2i - 1 - 2i + \gamma = -4$ $-2 + \gamma = -4$ $\gamma = -2$ Product of roots: $(-1 + 2i)(-1 - 2i) - 2 = -b$ $(1 + 4) - 2 = -b$ $b = 10$		C
6.  $r_2 = 2$ $r_1 = 2 - x$ $A_{\text{washer}} = \pi(r_2^2 - r_1^2)$ $= \pi(2^2 - (2 - x)^2)$ $= \pi(2^2 - (y^2)^2)$ $V = \int_0^{\sqrt{2}} \pi(2^2 - y^4) dy$		A
7. $\frac{x^2}{9} + y = 1$ Using $b^2 = a^2(1 - e^2)$ $1 = 9(1 - e^2)$ $\frac{1}{9} = 1 - e^2$ $e^2 = 1 - \frac{1}{9} = \frac{8}{9}$ $e = \frac{2\sqrt{2}}{3}$ Foci = $(\pm ae, 0)$ $= (\pm 3 \times \frac{2\sqrt{2}}{3}, 0)$ $= (\pm 2\sqrt{2}, 0)$		D

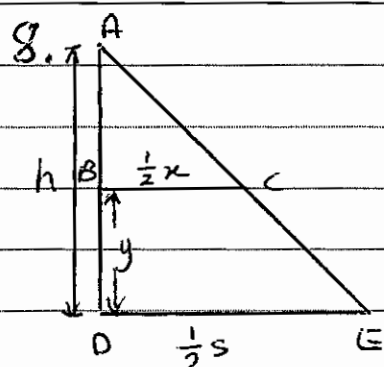
MATHEMATICS EXTENSION 2 – QUESTION

Multiple Choice.

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS



$$\text{In } \triangle ABC \parallel \triangle ADE$$

$$\frac{BC}{DE} = \frac{AB}{AD}$$

$$\frac{\frac{1}{2}x}{\frac{1}{2}s} = \frac{h-y}{h}$$

$$x = \frac{s(h-y)}{h}$$

C

$$9. \quad \ddot{x} = g - kv$$

$$\frac{dt}{dv} = \frac{1}{g - kv}$$

$$\int_0^t dt = \int_0^v \frac{1}{g - kv} dv$$

$$t = -\frac{1}{k} [\ln(g - kv) - \ln g]$$

$$-kt = \ln(g - kv) - \ln g$$

$$= \ln\left(\frac{g - kv}{g}\right)$$

$$e^{-kt} = \frac{g - kv}{g}$$

$$= 1 - \frac{kv}{g}$$

$$\frac{kv}{g} = 1 - e^{-kt}$$

$$v = \frac{g}{k} (1 - e^{-kt})$$

B

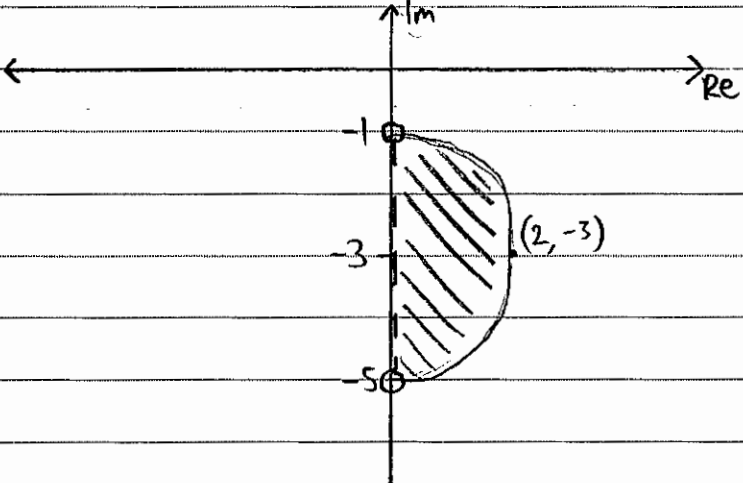
MATHEMATICS EXTENSION 2 – QUESTION Multiple Choice

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
10. $x = vt \cos \alpha$, $y = -\frac{1}{2}gt^2 + vt \sin \alpha$ $30 = 20t \cos \alpha$ $t = \frac{3}{2 \cos \alpha}$ — (1) $y = -\frac{gt^2}{2} + vt \sin \alpha$ $8.75 = -\frac{1}{2} \times 10 \times t^2 + 20t \sin \alpha$ $\times 4$ $35 = -20t^2 + 80t \sin \alpha$ — (2) Sub (1) in (2) $35 = -20 \left[\frac{3}{2 \cos \alpha} \right]^2 + 80 \left(\frac{3}{2 \cos \alpha} \right) \sin \alpha$ $= -45 \sec^2 \alpha + 120 \tan \alpha$ $7 = -9(1 + \tan^2 \alpha) + 24 \tan \alpha$ $9 \tan^2 \alpha - 24 \tan \alpha + 16 = 0$ $(3 \tan \alpha - 4)^2 = 0$ $3 \tan \alpha = 4$ $\tan \alpha = \frac{4}{3}$ $\alpha = \tan^{-1} \frac{4}{3}$		①.

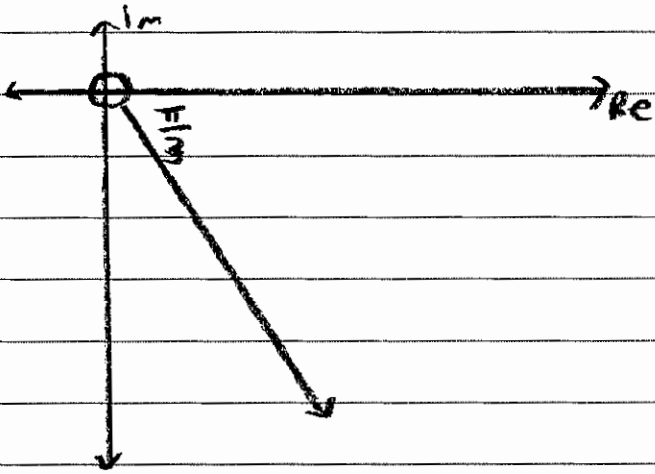
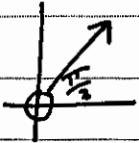
MATHEMATICS EXTENSION 2 – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
i $z = 2 \operatorname{cis} \frac{\pi}{3}$ $\therefore z^5 = 2^5 \operatorname{cis} \left(\frac{5\pi}{3} \right)$ $= 32 \operatorname{cis} \frac{5\pi}{3}$ or $32 \operatorname{cis} \left(-\frac{\pi}{3} \right)$ or $16 - 16\sqrt{3}i$	1	
ii $2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$ $= 1 + i\sqrt{3}$	1	
iii $\frac{1}{z} = \frac{1}{1 + i\sqrt{3}}$ $= \frac{1}{1 + i\sqrt{3}} \times \frac{1 - i\sqrt{3}}{1 - i\sqrt{3}}$ $= \frac{1 - i\sqrt{3}}{1 - 3i^2}$ $= \frac{1 - i\sqrt{3}}{4}$ $= \frac{1}{4} - \frac{i\sqrt{3}}{4}$	1	
iv $\omega^2 z = (2 - 3i)(2 - 3i)(1 + i\sqrt{3})$ $= (4 - 12i + 9i^2)(1 + i\sqrt{3})$ $= (-5 - 12i)(1 + i\sqrt{3})$ $= -5 - 12\sqrt{3}i^2 - 12i - 5\sqrt{3}i$ $= -5 + 12\sqrt{3} - (5\sqrt{3} + 12)i$	1	

MATHEMATICS EXTENSION 2 – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b) <u>i</u> let $u = x^2$ $\frac{du}{dx} = 2x$ $du = 2x dx$ $\int \frac{x}{1+x^4} dx = \frac{1}{2} \int \frac{2x dx}{1+(x^2)^2}$ $= \frac{1}{2} \int \frac{du}{1+u^2}$ $= \frac{1}{2} \tan^{-1} x^2 + C$	1 1	Everyone remembered to add "+C"! 😊
<u>ii</u> $\int \tan^3 x dx = \int \tan^2 x \tan x dx$ $= \int (\sec^2 x - 1) \tan x dx$ $= \int \sec^2 x \tan x dx - \int \tan x dx$ $= \int \sec^2 x \tan x dx + \int \frac{-\sin x}{\cos x} dx$ $= \frac{1}{2} \tan^2 x + \ln \cos x + C$	1 1	
c) <u>i</u> 	$\frac{1}{2}$ $\frac{1}{2}$	Essentially correct region All details (including broken line, open circles, labels on points)

MATHEMATICS EXTENSION 2 – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
9) ii 	1	 earned $\frac{1}{2}$ marks
d) $x^2 - xy + y^3 = 1$ Differentiating implicitly, $2x - (y + x \frac{dy}{dx}) + 3y^2 \frac{dy}{dx} = 0$ $3y^2 \frac{dy}{dx} - x \frac{dy}{dx} = y - 2x$ $\frac{dy}{dx} = \frac{y - 2x}{3y^2 - x}$	1	
At $P(1,1)$ $\frac{dy}{dx} = \frac{1 - 2}{3 - 1}$ $= -\frac{1}{2}$		
$\therefore$ gradient of tangent is $-\frac{1}{2}$ $\therefore$ gradient of normal is 2	1	
Equation of normal $y - 1 = 2(x - 1)$ $y = 2x - 2 + 1$ $\therefore y = 2x - 1$	1	

MATHEMATICS EXTENSION 2 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) $(x+1)^2$ is a factor of $Q(x)$ means $x = -1$ is a root of multiplicity 2 of $Q(x)$. $Q(x) = px^3 + 2x^2 + qx - 4$ $Q'(x) = 3px^2 + 4x + q$ Now $Q(-1) = 0$ and $Q'(-1) = 0$ So $Q(-1) = -p + 2 - q - 4 = 0$ $-p - q = 2$ $p + q = -2 \dots \textcircled{1}$		
$Q'(-1) = 3p - 4 + q = 0$ $3p + q = 4 \dots \textcircled{2}$	1	
$\textcircled{2} - \textcircled{1}$ $2p = 6$ $p = 3$ sub in $\textcircled{1}$	$\frac{1}{2}$	
$3 + q = -2$ $q = -5$	$\frac{1}{2}$	
$\therefore p = 3, q = -5$		
		$\textcircled{3}$

MATHEMATICS EXTENSION 2 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$b) i) \frac{x+1}{(x+3)(x+2)^2} = \frac{a}{x+3} + \frac{b}{x+2} + \frac{c}{(x+2)^2}$		
$x+1 = a(x+2)^2 + b(x+2)(x+3) + c(x+3)$		care needs to be taken when multiplying through by the LCD.
when $x = -2$: $-1 = 0 + 0 + c$ $c = -1$	1	Remember:
when $x = -3$: $-2 = a + 0 + 0$ $a = -2$	1	LCD = $(x+3)(x+2)^2$
Equating coef of x^2 : $0 = a + b$ $0 = -2 + b$ $b = 2$	1	
$\therefore a = -2, b = 2, c = -1$		
ii) $I = \int \frac{x+1}{(x+3)(x+2)^2} dx$		
$= \int \frac{-2}{x+3} dx + \int \frac{2}{x+2} dx - \int \frac{dx}{(x+2)^2}$		This part was done well
$= -2 \ln x+3 + 2 \ln x+2 - \int (x+2)^{-2} dx$		
$= 2 \left[\ln x+2 - \ln x+3 \right] - \frac{(x+2)^{-1}}{-1} + c$		
$= 2 \ln \left \frac{x+2}{x+3} \right + \frac{1}{x+2} + c$		(2)

MATHEMATICS EXTENSION 2 – QUESTION 12

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

c) Area of right-angled isosceles triangle:

$$A = \frac{1}{2}bh$$

$$A = \frac{1}{2}s^2 \quad \text{--- (1)}$$

Using Pythagoras Theorem:

$$s^2 + s^2 = (2y)^2$$

$$2s^2 = 4y^2$$

$$\frac{s^2}{2} = y^2 \quad \text{--- (2)}$$

but

$$x^2 + y^2 = 16$$

$$y^2 = 16 - x^2 \quad \text{--- (3)}$$

sub (2) in (3)

$$\frac{s^2}{2} = 16 - x^2$$

$$2s^2 = 2(16 - x^2) \quad \text{sub in (1)}$$

$$A = \frac{1}{2} [2(16 - x^2)]$$

$$= 16 - x^2$$

$$\delta V = (16 - x^2) \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-4}^4 (16 - x^2) \delta x$$

$$V = \int_{-4}^4 16 - x^2 dx$$

$$= 2 \times \int_0^4 16 - x^2 dx$$

$$= 2 \left[16x - \frac{x^3}{3} \right]_0^4$$

$$= 2 \left(\left[16 \times 4 - \frac{(4)^3}{3} \right] - 0 \right)$$

$$= \frac{256}{3} u^3$$

Care needs to be taken when finding the area of the right-angled isosceles triangle wrt x .

(4)

MATHEMATICS EXTENSION 2 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) $\vec{OA} = \frac{1}{2} \vec{OC}$ So $OA = \frac{1}{2} OC$ i.e. $OC = 2 OA$ but $ OA = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$ $= \sqrt{\frac{1}{4} + \frac{3}{4}}$ $= \sqrt{\frac{4}{4}}$ $= 1$ $\therefore OC = 2 \times 1$ $= 2$ and $\arg OA = \tan^{-1} \frac{\sqrt{3}/2}{-1/2}$ $= \tan^{-1} \sqrt{3}$ $= 2\pi/3$ but $\angle AOC = \pi/3$ $\therefore \angle COX = \frac{2\pi}{3} - \frac{\pi}{3}$ $= \frac{\pi}{3}$ So $C = 2 \operatorname{cis} \pi/3$ $= 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$ $= 2 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$ $= 1 + \sqrt{3} i$	1 1 1 1	Please refer to alternative method overleaf.

MATHEMATICS EXTENSION 2 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
Alternative solution 12 (d)		NOTE:
Also $\vec{OA} = \text{cis } \frac{2\pi}{3}$		Multiplying $C(z)$
and		through by $\text{cis } \theta$
$\vec{OC} = 2 \times \vec{OA} \times \text{cis } \left(-\frac{\pi}{3}\right)$		in an anticlockwise
$= 2 \times \text{cis } \frac{2\pi}{3} \times \text{cis } \left(-\frac{\pi}{3}\right)$		direction gives
$= 2 \text{cis } \left(\frac{2\pi}{3} + -\frac{\pi}{3}\right)$		$A(z)$
$= 2 \text{cis } \frac{\pi}{3}$		i.e
		$A(z) = C(z) \text{cis } \theta$
		So to find C
		we use
		$C(z) = A(z) \text{cis } (-\theta)$

MATHEMATICS EXTENSION 2 – QUESTION 13

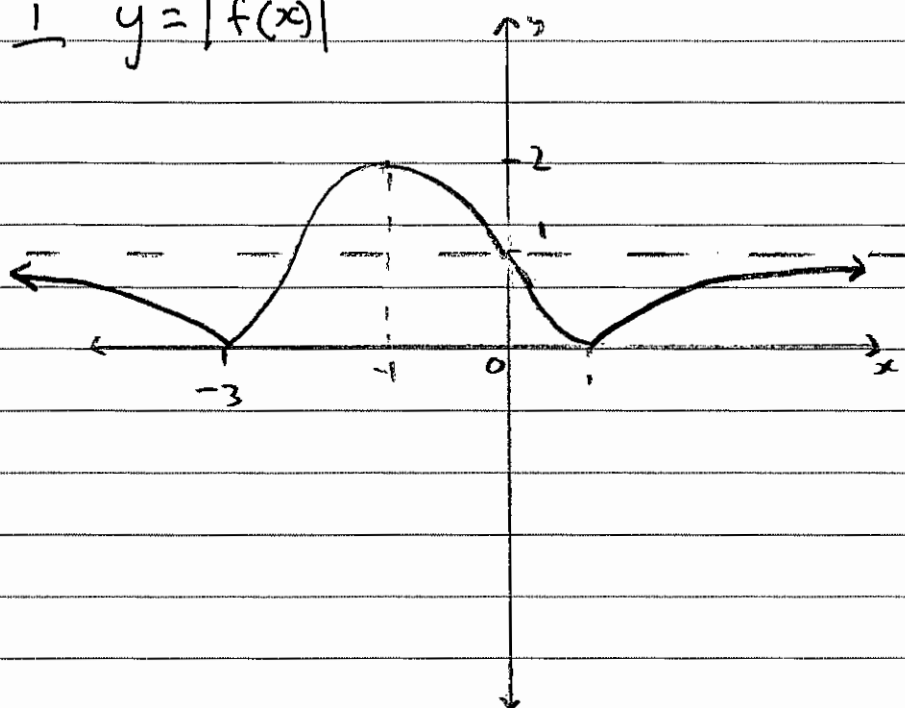
SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

a)

i $y = |f(x)|$



1

Crucial features:

- sharp point at x-intercepts
- asymptote at $y = 1$ (though no marks deducted this time)

Parts ii-iv were essentially:

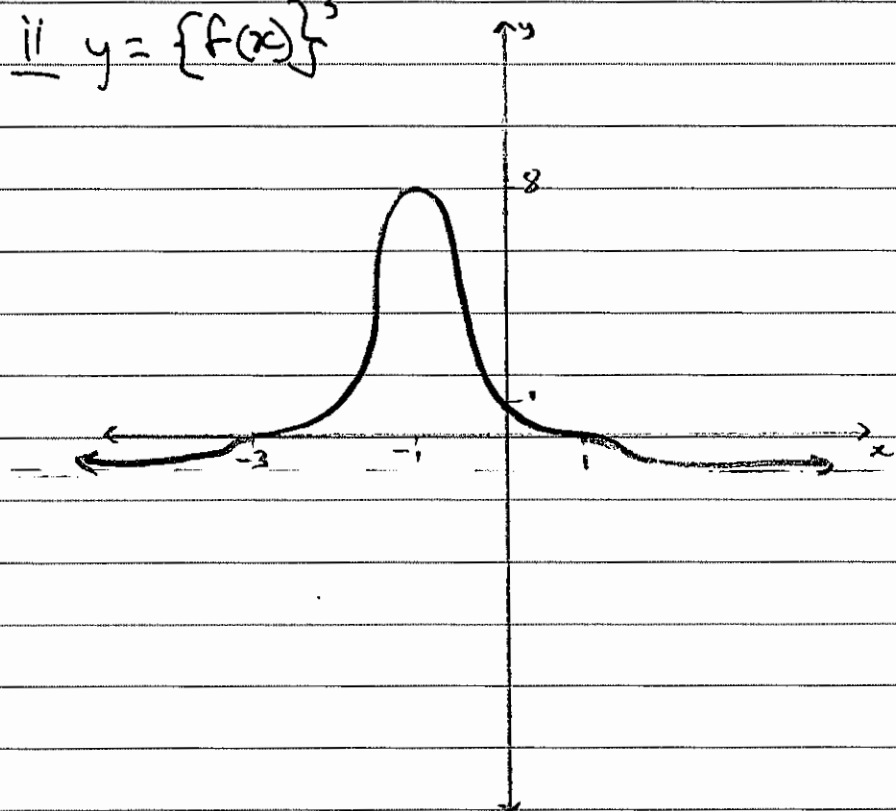
1

concept

1

execution

ii $y = \{f(x)\}^3$



1

Crucial features

- Horizontal inflexions at $x = -3$ and $x = 1$
- Maximum at $y = 8$
- Asymptote at $y = -1$
- Smooth, symmetric curve

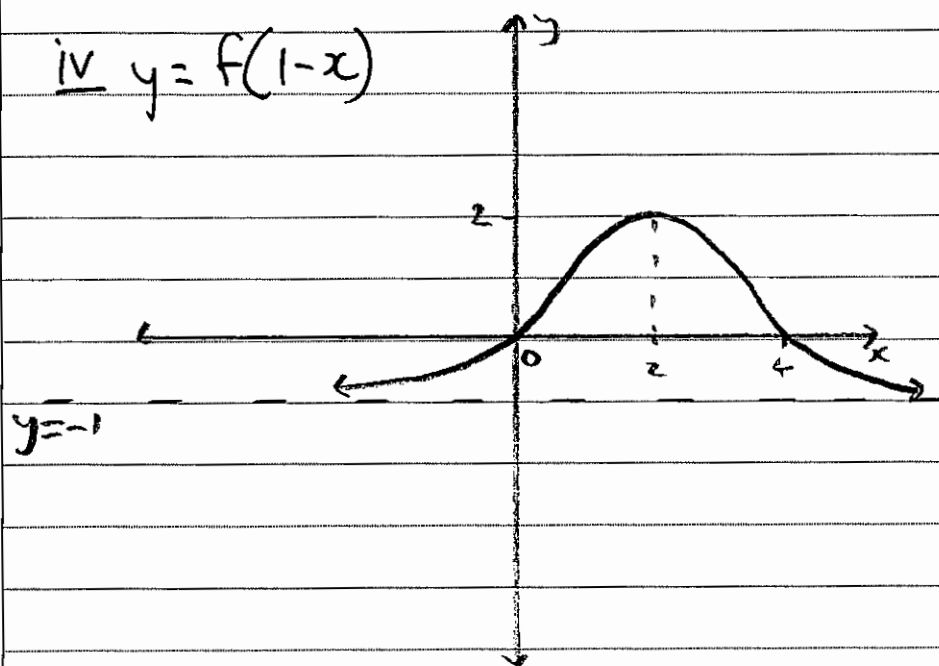
MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

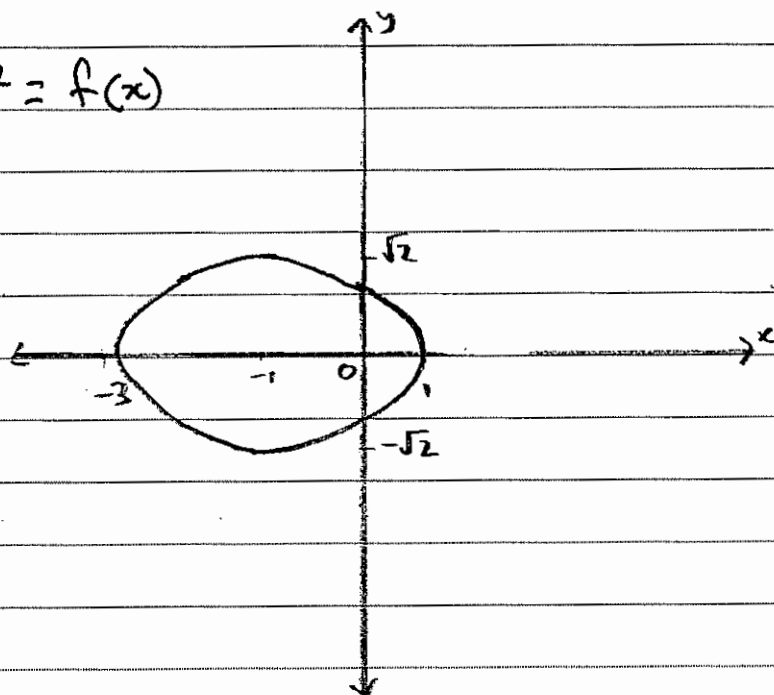
iv $y = f(1-x)$



- Crucial features:
- Intercepts at $x=0$ and $x=4$
 - Asymptote at $y=-1$
 - Maximum at $(2, 2)$

Note: part marks could be earned by building up your answer with a series of simpler transformations eg. sketching $y = f(-x)$ was helpful.

iii $y^2 = f(x)$



- Crucial features:
- Vertical at x -intercepts
 - Maximum at $y = \sqrt{2}$
 - Smooth, symmetric curve

MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b) Let $P(x) = x^4 - 6x^3 + 9x^2 + 4x - 12$ $P'(x) = 4x^3 - 18x^2 + 18x + 4$		
$P(2) = 16 - 48 + 36 + 8 - 12$ $= 0$		
$P'(2) = 32 - 72 + 36 + 4$ $= 0$		
$\therefore x=2$ is the double root $\therefore (x-2)^2$ is a factor	1	
$ \begin{array}{r} x^2 - 2x - 3 \\ x^2 - 4x + 4 \overline{) x^4 - 6x^3 + 9x^2 + 4x - 12} \\ \underline{x^4 - 4x^3 + 4x^2} \\ -2x^3 + 5x^2 + 4x \\ \underline{-2x^3 + 8x^2 - 8x} \\ -3x^2 + 12x - 12 \\ \underline{-3x^2 + 12x - 12} \\ 0 \end{array} $		
$\therefore x^4 - 6x^3 + 9x^2 + 4x - 12 = (x-2)^2 (x^2 - 2x - 3)$ $= (x-2)^2 (x-3)(x+1)$	1	Note that this question asked for a solution. Factorising was useful, but not enough to earn full marks.
$\therefore$ the solutions of $x^4 - 6x^3 + 9x^2 + 4x - 12 = 0$ are $x = -1, x = 2, x = 3$	1	
<u>OR</u> $x=2$ is the double root (as above)	1	
Let the other roots be α and β $\alpha + \beta + 2 + 2 = 6$ (sum of roots) $\alpha + \beta = 2$ $\alpha \times \beta \times 2 \times 2 = -12$ (product) $\alpha \beta = -3$	1	
$\therefore$ the roots are $-1, 3, 2$ and 2 $\therefore x = -1, x = 2, x = 3$	1	

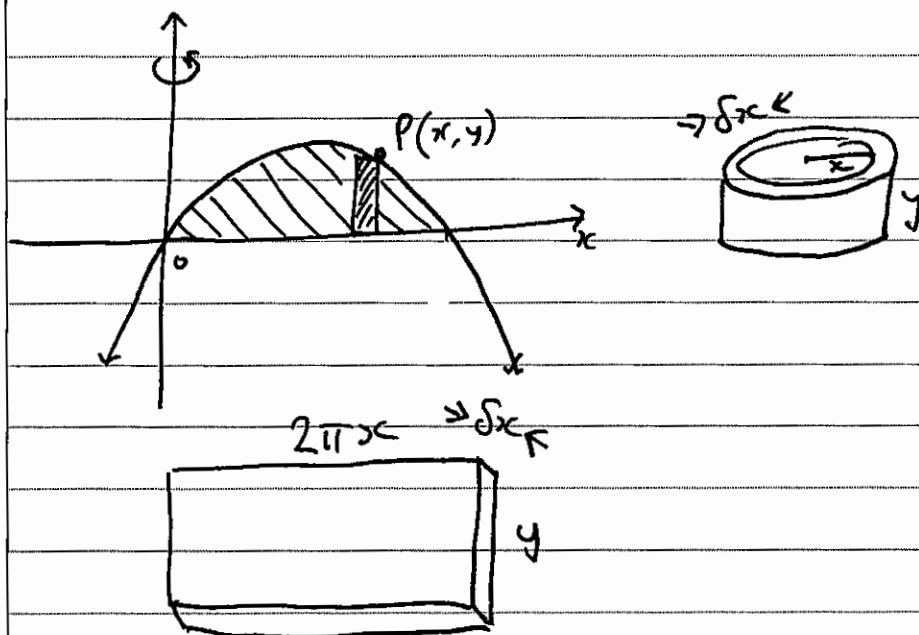
MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

c)



$$\begin{aligned} \therefore \delta V &= 2\pi x \cdot y \cdot \delta x \\ &= 2\pi x \cdot 4x(3-x) \cdot \delta x \end{aligned}$$

1

$$\begin{aligned} \therefore V &= \int_0^3 2\pi x \cdot 4x(3-x) dx \\ &= \int_0^3 24\pi x^2 - 8\pi x^3 dx \end{aligned}$$

1

$$\begin{aligned} &= \left[8\pi x^3 - 2\pi x^4 \right]_0^3 \\ &= (8\pi \cdot 27 - 2\pi \cdot 81) - (0 - 0) \end{aligned}$$

$$= 54\pi \text{ units}^3$$

1

MATHEMATICS EXTENSION 2 – QUESTION 13		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) If $x^3 - 5x - 2 = 0$ (1)		
Let $X = y + 1$		
$\therefore y = X - 1$		
sub $x-1$ into (1)		
$(x-1)^3 - 5(x-1) - 2 = 0$	1	
$x^3 - 3x^2 + 3x - 1 - 5x + 5 - 2 = 0$		
$x^3 - 3x^2 - 2x + 2 = 0$	1	
Note that this question requested an equation. Giving a function (e.g. $P(x) = x^3 - 3x^2 - 2x + 2$) as your answer does not answer the question.		

MATHEMATICS EXTENSION 2 – QUESTION 13		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) If $x^3 - 5x - 2 = 0$	(1)	
Let $X = y + 1$ $\therefore y = X - 1$ sub $x-1$ into (1)		
$(x-1)^3 - 5(x-1) - 2 = 0$	1	
$x^3 - 3x^2 + 3x - 1 - 5x + 5 - 2 = 0$		
$x^3 - 3x^2 - 2x + 2 = 0$	1	
Note that this question requested an equation. Giving a function (e.g. $P(x) = x^3 - 3x^2 - 2x + 2$) as your answer does not answer the question.		

[illegible]

MATHEMATICS EXTENSION 2 – QUESTION 13		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) If $x^3 - 5x - 2 = 0$	(1)	
Let $X = y + 1$ $\therefore y = X - 1$ sub $x-1$ into (1)		
$(x-1)^3 - 5(x-1) - 2 = 0$	1	
$x^3 - 3x^2 + 3x - 1 - 5x + 5 - 2 = 0$		
$x^3 - 3x^2 - 2x + 2 = 0$	1	
Note that this question requested an equation. Giving a function (e.g. $P(x) = x^3 - 3x^2 - 2x + 2$) as your answer does not answer the question.		

MATHEMATICS EXTENSION 2 – QUESTION 13		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) If $x^3 - 5x - 2 = 0$ (1)		
Let $X = y + 1$		
$\therefore y = X - 1$		
sub $x-1$ into (1)		
$(x-1)^3 - 5(x-1) - 2 = 0$	1	
$x^3 - 3x^2 + 3x - 1 - 5x + 5 - 2 = 0$		
$x^3 - 3x^2 - 2x + 2 = 0$	1	
Note that this question requested an equation. Giving a function (e.g. $P(x) = x^3 - 3x^2 - 2x + 2$) as your answer does not answer the question.		

MATHEMATICS EXTENSION 2 – QUESTION 13		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) If $x^3 - 5x - 2 = 0$ (1)		
Let $X = y + 1$		
$\therefore y = X - 1$		
sub $x-1$ into (1)		
$(x-1)^3 - 5(x-1) - 2 = 0$	1	
$x^3 - 3x^2 + 3x - 1 - 5x + 5 - 2 = 0$		
$x^3 - 3x^2 - 2x + 2 = 0$	1	
Note that this question requested an equation. Giving a function (e.g. $P(x) = x^3 - 3x^2 - 2x + 2$) as your answer does not answer the question.		

MATHEMATICS EXTENSION 2 – QUESTION 13		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) If $x^3 - 5x - 2 = 0$ (1)		
Let $X = y + 1$		
$\therefore y = X - 1$		
sub $x-1$ into (1)		
$(x-1)^3 - 5(x-1) - 2 = 0$	1	
$x^3 - 3x^2 + 3x - 1 - 5x + 5 - 2 = 0$		
$x^3 - 3x^2 - 2x + 2 = 0$	1	
Note that this question requested an equation. Giving a function (e.g. $P(x) = x^3 - 3x^2 - 2x + 2$) as your answer does not answer the question.		

MATHEMATICS EXTENSION 2 – QUESTION 13		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) If $x^3 - 5x - 2 = 0$ (1)		
Let $X = y + 1$		
$\therefore y = X - 1$		
sub $x-1$ into (1)		
$(x-1)^3 - 5(x-1) - 2 = 0$	1	
$x^3 - 3x^2 + 3x - 1 - 5x + 5 - 2 = 0$		
$x^3 - 3x^2 - 2x + 2 = 0$	1	
Note that this question requested an equation. Giving a function (e.g. $P(x) = x^3 - 3x^2 - 2x + 2$) as your answer does not answer the question.		

MATHEMATICS EXTENSION 2 – QUESTION 13		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) If $x^3 - 5x - 2 = 0$ (1)		
Let $X = y + 1$		
$\therefore y = X - 1$		
sub $x-1$ into (1)		
$(x-1)^3 - 5(x-1) - 2 = 0$	1	
$x^3 - 3x^2 + 3x - 1 - 5x + 5 - 2 = 0$		
$x^3 - 3x^2 - 2x + 2 = 0$	1	
Note that this question requested an equation. Giving a function (e.g. $P(x) = x^3 - 3x^2 - 2x + 2$) as your answer does not answer the question.		

MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

a) i) $P(2p, \frac{2}{p})$ $Q(2q, \frac{2}{q})$

Gradient of PQ: $m_{PQ} = \frac{\frac{2}{p} - \frac{2}{q}}{2p - 2q}$

$$= \frac{\frac{2q - 2p}{pq}}{2p - 2q}$$

$$= \frac{2q - 2p}{pq} \times \frac{1}{2(p - q)}$$

$$= \frac{-2(p - q)}{pq} \times \frac{1}{2(p - q)}$$

$$= -\frac{1}{pq}$$

$\frac{1}{2}$

Equation of chord PQ

$$y - \frac{2}{p} = -\frac{1}{pq}(x - 2p)$$

$$pqy - 2q = -x + 2p$$

$$\therefore x + pqy = 2(p + q)$$

} $\frac{1}{2}$

ii) For $xy = 4$.

Differentiating implicitly wrt x

$$y + x \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$= \frac{-2p}{-2p}$$

$$= -\frac{2}{p} \times \frac{1}{2p}$$

$$= -\frac{1}{p^2}$$

Part (a) i) ii) and
iii) was well
done.

①

MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

a) Cont'd

Equation of tangent:

$$y - \frac{2}{p} = -\frac{1}{p^2}(x - 2p)$$

$$p^2y - 2p = -x + 2p$$

$$x + p^2y = 4p \quad \dots \textcircled{1}$$

iii) Similarly tangent at Q:

$$x + q^2y = 4q \quad \dots \textcircled{2}$$

$$\textcircled{1} - \textcircled{2} \quad (p^2 - q^2)y = 4(p - q)$$

$$(p - q)(p + q)y = 4(p - q)$$

$$y = \frac{4}{p + q} \quad \text{sub in } \textcircled{1}$$

$$x + p^2\left(\frac{4}{p + q}\right) = 4p$$

$$x = 4p - \frac{4p^2}{p + q}$$

$$= \frac{4p(p + q) - 4p^2}{p + q}$$

$$= \frac{4p^2 + 4pq - 4p^2}{p + q}$$

$$= \frac{4pq}{p + q}$$

$$\therefore T \text{ is } \left(\frac{4pq}{p + q}, \frac{4}{p + q} \right)$$

2

2

MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) iv) Using T and Gradient of PQ, the line through T, parallel to PQ is: $y - \frac{4}{p+q} = -\frac{1}{pq} \left(x - \frac{4pq}{p+q} \right)$ Sub (0, 2) in this equation $2 - \frac{4}{p+q} = -\frac{1}{pq} \left(-\frac{4pq}{p+q} \right)$ $2(p+q) - 4 = 4$ $2(p+q) = 8$ $p+q = 4$	1 1	Some students did not substitute the point T and (0, 2) correctly in the equation of a line, otherwise the question was done quite well. (2)
b) Using Integration by parts $\int \frac{\ln x}{x^2} dx = uv - \int v u' dx$ $u = \ln x \quad v' = \frac{1}{x^2}$ $u' = \frac{1}{x} \quad v = x^{-2} = -\frac{1}{x}$ $= \ln x \cdot \frac{-1}{x} - \int -\frac{1}{x} \cdot \frac{1}{x} dx$ $= -\frac{\ln x}{x} + \int \frac{1}{x^2} dx$ $= -\frac{\ln x}{x} - \frac{1}{x} + c$	1 1	(2)

MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) i) $\cos 3A = \cos(2A + A)$ $= \cos 2A \cos A - \sin 2A \sin A$ $= (2\cos^2 A - 1)\cos A - 2\sin A \cos A \sin A$ $= 2\cos^3 A - \cos A - 2\cos A \sin^2 A$ $= 2\cos^3 A - \cos A - 2\cos A(1 - \cos^2 A)$ $= 2\cos^3 A - \cos A - 2\cos A + 2\cos^3 A$ $= 4\cos^3 A - 3\cos A$	1	
$\div 4$ $\frac{1}{4} \cos 3A = \cos^3 A - \frac{3}{4} \cos A \quad \dots \textcircled{1}$		
ii) For $x^3 - 6x + 2 = 0 \quad \dots \textcircled{2}$ Sub $x = 2\sqrt{2} \cos A$ in $\textcircled{2}$ $(2\sqrt{2} \cos A)^3 - 6(2\sqrt{2} \cos A) + 2 = 0$ $16\sqrt{2} \cos^3 A - 12\sqrt{2} \cos A + 2 = 0$ $16\sqrt{2} \cos^3 A - 12\sqrt{2} \cos A = -2$ $\cos^3 A - \frac{3}{4} \cos A = \frac{-2}{16\sqrt{2}}$ from $\textcircled{1}$ $\frac{1}{4} \cos 3A = \frac{-2}{16\sqrt{2}}$	1	
$\cos 3A = \frac{-1}{8\sqrt{2}} \times 4$ $= \frac{-1}{2\sqrt{2}}$		$\textcircled{2}$
iii) Three roots of $x^3 - 6x + 2 = 0$ are 3 roots of $\cos 3A = \frac{-1}{2\sqrt{2}}$ $3A = 2n\pi + \cos^{-1}\left(\frac{-1}{2\sqrt{2}}\right)$ $A = \frac{2n\pi}{3} + \cos^{-1}\left(\frac{-1}{2\sqrt{2}}\right)$ Take: $n=0, A_1 = \cos^{-1}\left(\frac{-1}{2\sqrt{2}}\right) \therefore x = 2\sqrt{2} \cos A_1 \doteq 2.268$ $n=1, A_2 = \frac{2\pi}{3} + \cos^{-1}\left(\frac{-1}{2\sqrt{2}}\right), x = 2\sqrt{2} \cos A_2 \doteq -2.26017$ $n=2, A_3 = \frac{4\pi}{3} + \cos^{-1}\left(\frac{-1}{2\sqrt{2}}\right), x = 2\sqrt{2} \cos A_3 \doteq 0.3398$ $\therefore x = 2.268, x = -2.6017, x = 0.3399$		1 mark to get 1 solution and $\frac{1}{2}$ mark each to get the other 2.
		$\textcircled{2}$

MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) Alternative solution to c(i) Using De Moivre's Theorem For $(\cos A + i \sin A)^3 = \cos 3A + i \sin 3A$ LHS = $\cos^3 A + 3 \cos^2 A i \sin A + 3 \cos A i^2 \sin^2 A + i^3 \sin^3 A$ $= \cos^3 A + 3 \cos A \sin A i - 3 \cos A \sin^2 A - i \sin^3 A$ Equating Real Parts $\cos 3A = \cos^3 A - 3 \cos A \sin^2 A$ $= \cos^3 A - 3 \cos A (1 - \cos^2 A)$ $= \cos^3 A - 3 \cos A + 3 \cos^3 A$ $= 4 \cos^3 A - 3 \cos A$ $\therefore$ $\frac{1}{4} \cos 3A = \cos^3 A - \frac{3}{4} \cos A$		

MATHEMATICS EXTENSION 2 – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) Let $t = \tan \frac{\theta}{2}$ $\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$ $= \frac{1 + \tan^2 \frac{\theta}{2}}{2} \quad (\text{since } \sec^2 \theta = 1 + \tan^2 \theta)$ $= \frac{1 + t^2}{2}$ $\therefore \frac{d\theta}{dt} = \frac{2}{1 + t^2}$ $d\theta = \frac{2 dt}{1 + t^2}$	1	This is a "show" question - don't take shortcuts. <u>Show</u> how you obtain every result.
<u>OR</u> $t = \tan \frac{\theta}{2}$ $\tan^{-1} t = \frac{\theta}{2}$ $\theta = 2 \tan^{-1} t$ $\frac{d\theta}{dt} = \frac{2}{1 + t^2}$ $d\theta = \frac{2 dt}{1 + t^2}$	1	When doing integration by substitution, never mix variables.
when $\theta = 0$, $t = \tan \frac{\theta}{2} = 0$		Your integral should progress from one wholly in terms of θ, to one wholly in terms of t.
when $\theta = \frac{\pi}{3}$, $t = \tan \frac{\pi/3}{2}$ $= \tan \frac{\pi}{6}$ $= \frac{1}{\sqrt{3}}$		
$\therefore \int_0^{\frac{\pi}{3}} \sec \theta d\theta = \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{\cos \theta} d\theta$ $= \int_0^{\frac{1}{\sqrt{3}}} \frac{1 + t^2}{1 - t^2} \cdot \frac{2 dt}{1 + t^2}$	1	

MATHEMATICS EXTENSION 2 – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2}{1-t^2} dt$ $\text{Let } \frac{2}{(1-t)(1+t)} \equiv \frac{a}{1-t} + \frac{b}{1+t}$ $\therefore 2 = a(1+t) + b(1-t)$ $\text{when } t=1, 2 = 2a$ $\therefore a=1$ $\text{when } t=-1, 2 = 2b$ $b=1$		
$= \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1+t} + \frac{1}{1-t} dt$		
$= \left[\ln 1+t - \ln 1-t \right]_0^{\frac{1}{\sqrt{3}}}$		
$= \left[\ln \frac{ 1+t }{ 1-t } \right]_0^{\frac{1}{\sqrt{3}}}$		
$= \ln \left(\frac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}} \right)$		
$= \ln \left(\frac{\sqrt{3}+1}{\sqrt{3}-1} \right)$		
$= \ln \frac{(\sqrt{3}+1)(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)}$		
$= \ln \left(\frac{4+2\sqrt{3}}{3-1} \right)$	1	for full solution
$= \ln(2+\sqrt{3})$		

MATHEMATICS EXTENSION 2 – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b) i) $I_n = \int_1^e (1 - \ln x)^n dx, n = 0, 1, 2, \dots$ let $u = (1 - \ln x)^n$ $\therefore u' = -\frac{1}{x} \cdot n(1 - \ln x)^{n-1}$ let $v' = 1$ $\therefore v = x$ $\therefore I_n = \left[x(1 - \ln x)^n \right]_1^e - \int_1^e x \cdot -\frac{1}{x} \cdot n(1 - \ln x)^{n-1} dx$ $= \left[e(1-1) - 1(1-0) \right] + n \int_1^e (1 - \ln x)^{n-1} dx$ $= -1 + n I_{n-1}$		
ii) $I_3 = -1 + 3I_2$ $= -1 + 3(-1 + 2I_1)$ $= -1 + 3(-1 + 2(-1 + I_0))$ $= -1 + 3(-1 + 2(-1 + (e-1)))$ $= -1 + 3(-1 + 2(-1 + e - 1))$ $= -1 + 3(-1 + 2(e - 2))$ $= -1 + 3(-1 + 2e - 4)$ $= -1 + 3(2e - 5)$ $= -1 + 6e - 15$ $= 6e - 16$	$I_0 = \int_1^e (1 - \ln x)^0 dx$ $= \int_1^e 1 dx$ $= [x]_1^e$ $= e - 1$	2
2 marks for full solution		
1 mark for significant progress towards solution		

MATHEMATICS EXTENSION 2 – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
Q1 i $9x^2 - 16y^2 = 144$ $\therefore \frac{x^2}{16} - \frac{y^2}{9} = 1$ $a^2 = 16$ $b^2 = 9$ $a = 4$ $b = 3$ and $b^2 = a^2(e^2 - 1)$ $9 = 16(e^2 - 1)$ $\frac{9}{16} = e^2 - 1$ $e^2 = \frac{25}{16}$ $e = \frac{5}{4}$ ($e > 0$) Foci: $(\pm ae, 0)$ $(\pm 4 \times \frac{5}{4}, 0)$ $\therefore S(5, 0), S'(-5, 0)$	1	
ii $\frac{x^2}{16} - \frac{y^2}{9} = 1$ $\frac{2x}{16} - \frac{2y}{9} \cdot \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{2x}{16} \times \frac{9}{2y}$ $= \frac{9x}{16y}$ at $P(x_1, y_1)$, $m_T = \frac{9x_1}{16y_1}$ $\therefore$ equation of tangent to H at P $y - y_1 = \frac{9x_1}{16y_1}(x - x_1)$ $16y_1y - 16y_1^2 = 9x_1x - 9x_1^2$	1	

MATHEMATICS EXTENSION 2 – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$\therefore 9x, x - 16y, y = 9x,^2 - 16y,^2$ $= 144 \text{ (since } P(x, y) \text{ lies on } H)$	1	
$\therefore 9x, x - 16y, y = 144$		
iii when $y = 0$ $9x, x - 16y, (0) = 144$ $x = 144$ $\frac{9x,}{= \frac{16}{x,}}$ $\therefore G\left(\frac{16}{x,}, 0\right)$	1	
iv		$9x^2 - 16y^2 = 144$ $\Rightarrow y^2 = \frac{9x^2 - 144}{16}$
$SP^2 = (x, -5)^2 + (y, -0)^2$ $= x,^2 - 10x, + 25 + y,^2$ $= x,^2 - 10x, + 25 + \frac{9x,^2 - 144}{16} \text{ (since } P(x, y) \text{ lies on } H)$ $= \frac{16x,^2 - 160x, + 400 + 9x,^2 - 144}{16}$ $= \frac{25x,^2 - 160x, - 256}{16}$		

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SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$= \frac{(5x-16)^2}{4^2}$		
$\therefore SP = \frac{5x-16}{4}$	1	
$S'P^2 = (x, -5)^2 + (y, -0)^2$ $= x^2 + 10x + 25 + y^2$ $= x^2 + 10x + 25 + \frac{9x^2 - 144}{16}$ $= \frac{16x^2 + 160x + 400 + 9x^2 - 144}{16}$ $= \frac{25x^2 + 160x - 256}{16}$ $= \frac{(5x+16)^2}{4^2}$		
$\therefore S'P = \frac{5x+16}{4}$	1	for any other distance
$\frac{SP}{S'P} = \frac{(5x-16)/4}{(5x+16)/4}$ $= \frac{(5x-16)}{(5x+16)}$		(i.e. S'P, SG, or S'G)
$SG = 5 - \frac{16}{x}$ $= \frac{5x-16}{x}$		
$S'G = 5 + \frac{16}{x}$ $= \frac{5x+16}{x}$		
$\therefore \frac{SG}{S'G} = \frac{(5x-16)/x}{(5x+16)/x}$ $= \frac{5x-16}{5x+16}$ $= \frac{SP}{S'P}$	1	full solution

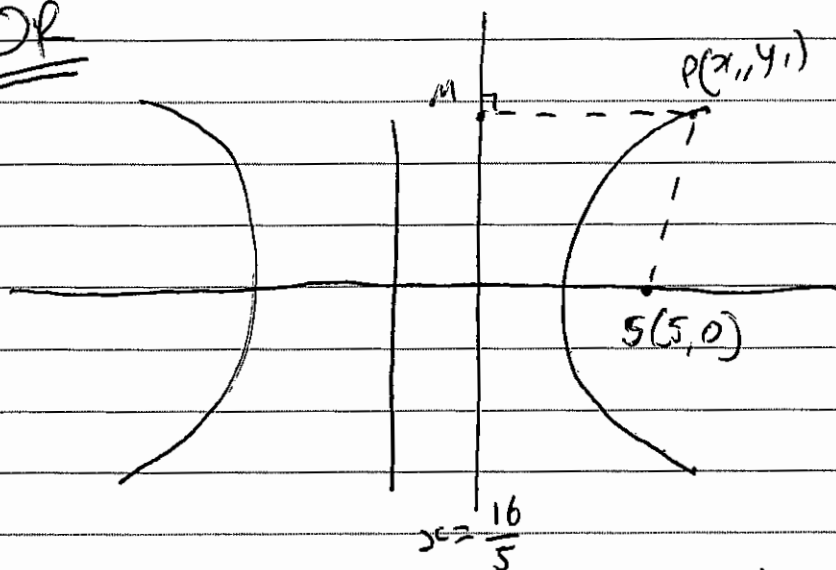
MATHEMATICS EXTENSION 2 – QUESTION 15

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

OP



directrix has equation $x = \frac{a}{e} \Rightarrow \frac{4}{5/4}$
 $\therefore x = \frac{16}{5}$

Let M be the foot of the perpendicular from the directrix to P

$$\therefore M\left(\frac{16}{5}, y\right)$$

From the definition of a hyperbola:

$$SP = e PM$$

$$= \frac{5}{4} \left(x, -\frac{16}{5}\right)$$

$$= \frac{5}{4} \left(\frac{5x}{5}, \frac{-16}{5}\right)$$

$$= \frac{5x, -16}{4}$$

Similarly, $SP = e PM'$

$$= \frac{5}{4} \left(x, -\frac{16}{5}\right)$$

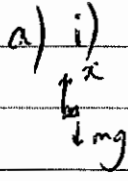
$$= \frac{5}{4} \left(x, +\frac{16}{5}\right)$$

$$= \frac{5}{4} \left(\frac{5x}{5}, \frac{+16}{5}\right)$$

$$= \frac{5x, +16}{4}$$

Then as above.

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) i)  $m\ddot{x} = -mg - R$ $= -mg - \frac{mgv^2}{a^2}$ $\ddot{x} = -g - \frac{gv^2}{a^2} \quad \text{but } \ddot{x} = v \cdot \frac{dv}{dx}$ $v \cdot \frac{dv}{dx} = -g \left(1 + \frac{v^2}{a^2} \right)$ $= -g \left(\frac{a^2 + v^2}{a^2} \right)$ $= -\frac{g}{a^2} (a^2 + v^2)$	1 1	
ii) $\frac{dv}{dx} = -\frac{g}{a^2} \left(\frac{a^2 + v^2}{v} \right)$ $\frac{dx}{dv} = -\frac{a^2}{g} \left(\frac{v}{a^2 + v^2} \right)$ $x = -\frac{a^2}{g} \int \frac{v}{a^2 + v^2} dv$ $= -\frac{a^2}{2g} \ln(a^2 + v^2) + c$ when $t=0$, $x=0$, $v=U$ $0 = -\frac{a^2}{2g} \ln(a^2 + U^2) + c$ $c = \frac{a^2}{2g} \ln(a^2 + U^2)$ $x = -\frac{a^2}{2g} \ln(a^2 + v^2) + \frac{a^2}{2g} \ln(a^2 + U^2)$ $= \frac{a^2}{2g} \ln \left[\frac{a^2 + U^2}{a^2 + v^2} \right]$ For greatest height, $v=0$ $x_{\max} = \frac{a^2}{2g} \ln \left[\frac{a^2 + U^2}{a^2} \right]$ $= \frac{a^2}{2g} \ln \left[1 + \frac{U^2}{a^2} \right]$	1/2 1/2 1	

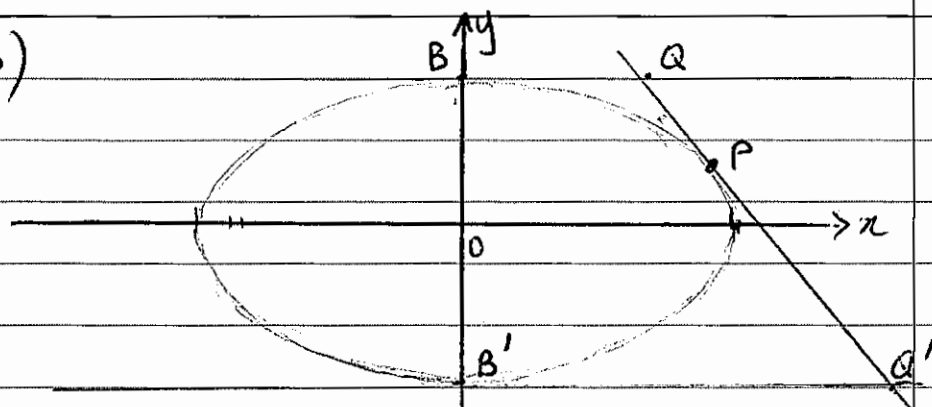
MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

b)



The equation of the tangent at P

$$bx \cos \theta + ay \sin \theta - ab = 0 \quad \text{--- (1)}$$

Sub $y = b$ in (1) to obtain x coord of Q.

$$bx \cos \theta + ab \sin \theta - ab = 0$$

$$bx \cos \theta = ab - ab \sin \theta$$

$$x = \frac{ab(1 - \sin \theta)}{b \cos \theta}$$

$$= \frac{a(1 - \sin \theta)}{\cos \theta}$$

$\frac{1}{2}$

$$\therefore BQ = \frac{a(1 - \sin \theta)}{\cos \theta}$$

$\frac{1}{2}$

Sub $y = -b$ in (1) to obtain x -coord of Q'

$$bx \cos \theta - ab \sin \theta - ab = 0$$

$$x = \frac{ab(1 + \sin \theta)}{b \cos \theta}$$

$$x = \frac{a(1 + \sin \theta)}{\cos \theta}$$

$\frac{1}{2}$

$$\therefore BQ' = \frac{a(1 + \sin \theta)}{\cos \theta}$$

$\frac{1}{2}$

$$\text{Now } BQ \times BQ' = \frac{a(1 - \sin \theta)}{\cos \theta} \times \frac{a(1 + \sin \theta)}{\cos \theta}$$

1

$$= \frac{a^2(1 - \sin^2 \theta)}{\cos^2 \theta}$$

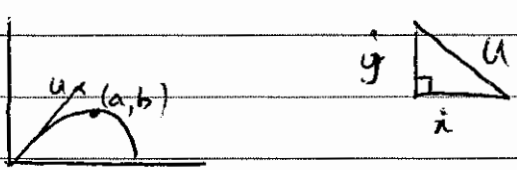
$$= \frac{a^2 \cos^2 \theta}{\cos^2 \theta}$$

$$= a^2$$

} 1

4

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c)  Initial Velocities $\dot{x} = u \cos \theta$ $\dot{y} = u \sin \theta$		
Vertical Component $\ddot{y} = -g$ $\dot{y} = -gt + c$ When $t=0$, $\dot{y} = u \sin \theta$ $u \sin \theta = 0 + c$ $\therefore \dot{y} = -gt + u \sin \theta$ $y = -\frac{gt^2}{2} + ut \sin \theta + c_2$ --- (1) When $t=0$, $y=0$, $0 = 0 + 0 + c_2 \therefore c_2 = 0$ $y = -\frac{1}{2}gt^2 + ut \sin \theta$	Horizontal Component $\ddot{x} = 0$ $\dot{x} = c_3$ When $t=0$, $\dot{x} = u \cos \theta$ $\dot{x} = u \cos \theta$ $x = ut \cos \theta + c_4$ When $x=0$, $t=0$ $0 = 0 + c_4$ $c_4 = 0$ $x = ut \cos \theta$ --- (2) $t = \frac{x}{u \cos \theta}$ --- (3)	2 marks for fully deriving the vertical and horizontal equations of motion 1 mark for only stating them.
Sub (3) in (1) $y = -\frac{1}{2}g\left(\frac{x}{u \cos \theta}\right)^2 + u\left(\frac{x}{u \cos \theta}\right) \sin \theta$ $= -\frac{1}{2}g\left(\frac{x^2}{u^2 \cos^2 \theta}\right) + \frac{x \sin \theta}{\cos \theta}$ $y = -\frac{1}{2}g \frac{x^2}{u^2} \sec^2 \theta + x \tan \theta$ --- (4)	2 1	
		(3)

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) ii)		
From (4)		
$y = \frac{-g}{2u^2} x^2 (1 + \tan^2 \theta) + x \tan \theta \dots (5)$		
when $x=a$, $y=b$ sub in (5)		
$b = \frac{-g}{2u^2} a^2 (1 + \tan^2 \theta) + a \tan \theta$	1	
$b = \frac{-ga^2}{2u^2} - \frac{ga^2 \tan^2 \theta}{2u^2} + a \tan \theta$		
$\frac{ga^2 \tan^2 \theta}{2u^2} - a \tan \theta + \frac{ga^2}{2u^2} + b = 0$		
$ga^2 \tan^2 \theta - 2u^2 a \tan \theta + ga^2 + 2u^2 b = 0$		
Quad Equation in $\tan \theta$		
For 2 distinct points, $\Delta > 0$		
$b^2 - 4ac > 0$		
$(-2u^2 a)^2 - 4(ga^2)(ga^2 + 2u^2 b) > 0$	1	
$4u^4 a^2 - 4(g^2 a^4 + 2ga^2 b u^2) > 0$		
$4u^4 a^2 - 8ga^2 b u^2 - 4g^2 a^4 > 0$		
$\div 4a^2 \quad u^4 - 2gbu^2 - g^2 a^2 > 0$		
$u^4 - 2gbu^2 > g^2 a^2$		
By Completing the square on LHS	1	
$u^4 - 2gbu^2 + g^2 b^2 > g^2 a^2 + g^2 b^2$		
$(u^2 - gb)^2 > g^2(a^2 + b^2)$		
		(3)